

Banking Campaign Optimization & Profit Strategy Report

1. Executive Summary

This report analyzes a bank's telemarketing campaign data (45,211 customers) to identify high-performing customer segments, optimize contact strategy, and maximize profitability under budget constraints.

Key findings show that data-driven targeting significantly improves campaign performance. By focusing on high-probability segments, conversion rates can increase from 11.7% to over 20%, delivering up to 79% higher profit without increasing budget.

2. Business Problem

The bank faces several challenges:

- High cost per call
- Low overall conversion rate
- Limited marketing budget
- Inefficient targeting strategy

Objective:

- Identify customers most likely to subscribe
 - Optimize campaign frequency
 - Improve profitability using data-driven decisions
-

3. Data Overview

Dataset includes:

- Customer demographics (age, job, education)
- Financial indicators (balance, loan status)
- Campaign interactions (duration, campaign count)
- Outcome variable (subscription: yes/no)

Total records analyzed: 45,211

4. Data Cleaning & Preparation

Steps performed:

- Standardized categorical values (lowercase, trimmed)
- Removed invalid durations
- Converted invalid pdays (-1) to NULL
- Checked for missing values

Result: Clean, analysis-ready dataset

5. Overall Campaign Performance

- Total Customers Contacted: 45,211
- Conversion Rate: 11.7%
- Total Profit: €831,745

Insight:

The campaign is profitable but inefficient, with a large number of unsuccessful calls.

6. Customer Insights

High Performing Segments

- Students: ~28.7% conversion
- Retirees: ~22.8%
- High balance customers (>€1000)

Low Performing Segments

- Blue-collar workers: ~7%
- Entrepreneurs: ~8%

Key Insight

Higher financial stability and certain occupations significantly increase likelihood of subscription.

7. Contact Strategy Analysis

Campaign Frequency

- 1 call: ~14.6% conversion

- 2–3 calls: ~11%
- 4+ calls: <9%

Insight:

Over-contacting reduces effectiveness due to customer fatigue.

Call Duration

- Longer calls (>500 sec): ~42% conversion
- Short calls (<100 sec): ~1%

Insight:

Call duration strongly correlates with interest, but cannot be used for pre-targeting.

8. Timing Strategy

Top months:

- March (~52%)
- September–December (~44–47%)

Worst month:

- May (~6.7%)

Insight:

Campaign timing significantly impacts success rates.

9. Risk Profiling

Key Risk Segments

- High balance customers not subscribing
- Low balance customers with low ROI
- Heavily contacted but unsuccessful customers

Insight:

Repeated calls to low-probability customers waste budget.

10. Profit Optimization

Historical Profit

- Net Profit: €831,745

Targeted Strategy

Conditions:

- Balance > €1363
- Campaign ≤ 3
- Loan = No
- Previous success = Yes

Results:

- Conversion Rate: ~64%
- Net Profit: ~€63,000

Insight:

Targeted campaigns drastically improve efficiency.

11. Budget-Constrained Strategy (20,000 Calls)

Approaches Compared

1. Mass Calling
 - Highest total profit
 - Inefficient resource usage
2. Targeted Calling
 - Very high conversion (~64%)
 - Limited scale
3. Ranked Strategy (Best)
 - Balanced approach
 - Conversion ~16%
 - Profit ~€543,840

Insight:

Ranking customers using data-driven scoring provides optimal balance between scale and efficiency.

12. Scenario Analysis

Scenarios tested:

- Conservative (low conversion, high cost)
- Base case (current performance)
- Optimistic (improved efficiency)
- Best segment (targeted strategy)

Key Insight:

Targeted campaigns deliver highest ROI (up to 474%).

13. Strategic Recommendations

Target

- Students and retirees
- High balance customers
- Customers with previous success

Avoid

- Blue-collar and low conversion segments
- Over-contacted customers

Campaign Strategy

- Limit to 1–2 calls per customer

Timing

- Focus campaigns in March and Q4

Expected Impact

- Conversion increase: 11.7% → 20%+
 - Profit increase: ~79%
-

14. Conclusion

Data-driven targeting significantly improves campaign efficiency and profitability, delivering higher ROI while reducing unnecessary marketing costs compared to mass outreach strategies.

15. Tools Used

- MySQL (Data analysis)
- Microsoft Excel (Dashboard & Simulation)

16. Final Recommendation

If implemented, the bank should shift from mass calling to a targeted, data-driven strategy focusing on high-value customers, optimized timing, and controlled contact frequency to maximize return on investment.